Paper 1

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1 Introduction

2 Formal Compression–Recovery Framework

We formalize the structural conditions under which information can be recovered after extreme scrambling, without requiring preservation of full microstate data.

2.1 State Space and Scrambling

Let $\mathcal{X}$ be a state space with microstates $x \in \mathcal{X}$.

Let $\{\mathcal{A}_\mu\}_{\mu \in \mathcal{S}}$ be a partition of $\mathcal{X}$ into attractor basins (or sectors), indexed by μ in a finite set $\mathcal{S}$.

Each basin corresponds to a coarse-grained identity class or sector.

Define a scrambling map

$$\mathcal{S} : \mathcal{X} \to \mathcal{X}$$

such that local observables on $\mathcal{S}(x)$ approach maximum entropy (thermal appearance), while global invariants may remain correlated with the original basin.

We assume scrambling destroys fine-grained information but does not necessarily eliminate coarse-grained sector structure.

2.2 Sector Map

Define a sector map

$$K : \mathcal{X} \to \mathcal{K}$$

where $\mathcal{K}$ is a finite set of sector labels.

We require that K is constant on attractor basins:

$$x \in \mathcal{A}_\mu \quad \Rightarrow \quad K(x) = \mu.$$

Thus, K identifies basin membership but does not encode microstate detail.

Let

$$M_{sectors} = |\mathcal{K}|.$$

Define the side-information complexity:

$$k := \log_2 M_{sectors}.$$

2.3 Decoder Flow

Let $\mathcal{D}_\mu : \mathcal{X} \to \mathcal{X}$ be a basin-attracting flow associated with sector μ.

We define recovery as the dynamical process:

$$x\mathcal{S}\tilde{x} \quad , \quad K(x) \textit{ retained} \quad , \quad \tilde{x}\mathcal{D}_{K(x)}\hat{x} \in \mathcal{A}_{K(x)}.$$

Recovery succeeds if

$$\hat{x} \in \mathcal{A}_{K(x)}.$$

We do not require $\hat{x} = x$ (no cloning). Only basin equivalence is required.

2.4 Compression–Recovery Theorem

[Compression–Recovery] If recovery after scrambling is possible via basin selection, then the minimal side-information required satisfies

$$k \sim \log_2 M_{sectors}.$$

[Sketch] Recovery requires distinguishing which of the $M_{sectors}$ basins the original state belonged to.

Any labeling scheme that uniquely identifies basin membership must encode at least $\log_2 M_{sectors}$ bits.

If $M_{sectors}$ grows subexponentially relative to microstate entropy S, then $k \ll S$ and compression is possible.

If $M_{sectors} \sim e^S$, then $k \sim S$ and no nontrivial compression exists.

2.5 Thermal Appearance vs. Correlation Preservation

Let O_{local} be a set of local observables.

Scrambling may produce:

$$S_{local}(\mathcal{S}(x)) \approx S_{\max}$$

indicating thermal appearance.

However, mutual information

$$I(K(x); \mathcal{S}(x))$$

may remain nonzero if global correlations preserve sector information.

Recovery requires:

$$I(K(x); \mathcal{S}(x)) > 0.$$

Thus, apparent thermality does not imply sector information loss.

3 Hopfield Network Instantiation

We now instantiate the compression–recovery framework in a classical Hopfield network, providing a concrete, analyzable example where recovery after scrambling is possible with logarithmic side information.

3.1 Model Definition

Consider a Hopfield network with N binary neurons

$$s_i \in \{-1, +1\}, \quad i = 1, \ldots, N.$$

Let $\{\xi^\mu\}_{\mu=1}^P$ be P stored patterns, with

$$\xi_i^\mu \in \{-1, +1\}$$

drawn independently with equal probability.

The synaptic weight matrix is defined as

$$W_{ij} = \frac{1}{N} \sum_{\mu=1}^{P} \xi_i^\mu \xi_j^\mu, \quad W_{ii} = 0.$$

The network energy is

$$E(s) = -\frac{1}{2} \sum_{i \neq j} W_{ij} s_i s_j.$$

Each stored pattern ξ^μ corresponds to an attractor basin $\mathcal{A}_\mu$ in state space.

3.2 Attractor Basins as Sectors

Define the overlap between a state s and pattern ξ^μ:

$$m^\mu(s) = \frac{1}{N} \sum_{i=1}^{N} s_i \xi_i^\mu.$$

A state belongs to basin $\mathcal{A}_\mu$ if

$$m^\mu(s) > m^\nu(s) \quad \forall \nu \neq \mu.$$

Thus, the attractor index μ defines a sector label:

$$K(s) := \arg\max_\mu m^\mu(s).$$

The number of sectors is

$$M_{sectors} = P.$$

The side-information complexity is therefore

$$k = \log_2 P.$$

3.3 Scrambling

Define a scrambling operation $\mathcal{S}$ as random bit flips:

$$s_i \mapsto \{ s_i \, withprobability 1-q, \, -s_i withprobability q,$$

with $q \in (0, 1/2)$.

For sufficiently large q, local observables approach maximal entropy:

$$E[s_i] \approx 0, \quad E[s_i s_j] \approx 0 \quad (i \neq j),$$

and the scrambled state appears thermal.

However, for q below the basin destruction threshold, the ordering of overlaps $\{m^\mu\}$ remains correlated with the original basin index μ.

3.4 Recovery Dynamics

Recovery is performed by standard asynchronous Hopfield updates:

$$s_i(t+1) = sign\left(\sum_j W_{ij} s_j(t)\right).$$

Let $s_0 \in \mathcal{A}_\mu$ be an initial state, $\tilde{s} = \mathcal{S}(s_0)$ the scrambled state, and $\hat{s}$ the fixed point reached under dynamics.

Recovery succeeds if

$$\hat{s} \in \mathcal{A}_\mu.$$

Exact microstate recovery is not required. Only basin recovery is demanded.

3.5 Information Scaling

The full microstate entropy scales as

$$S_{micro} \sim N.$$

However, recovery requires only identification of the basin index μ, which requires

$$k = \log_2 P$$

bits of side information.

As long as

$$P \ll e^{\alpha N}$$

for any $\alpha > 0$, we have

$$k \ll S_{micro},$$

demonstrating exponential compression.

This realizes the Compression–Recovery Theorem explicitly.

3.6 Failure Regime

If P approaches the storage capacity

$$P_c \approx 0.138\, N,$$

basins shrink and eventually disappear.

In this regime:

- $M_{sectors} \sim e^{cN}$,
- $k \sim N$,
- compression fails,
- recovery becomes impossible.

This matches the theorem's prediction that recovery requires subexponential sector growth.

3.7 Interpretation

The Hopfield network demonstrates:

1. Scrambling can destroy local structure (thermal appearance).
2. Global correlations (overlaps) preserve sector information.
3. Recovery requires only $\log P$ bits.
4. Exact state cloning is neither possible nor required.

Thus, Hopfield dynamics provide a concrete, classical realization of compression–recovery after scrambling.

4 Black Hole Information: A Conditional Prediction from Compression–Recovery

4.1 The information paradox as a scrambling–recovery question

The black hole information paradox arises from the tension between (i) quantum mechanical unitarity and (ii) Hawking's semiclassical result that emitted radiation is locally thermal. If evaporation maps a pure initial state to a thermal mixed state with no recoverable correlations, information would be destroyed. We do not claim to resolve this paradox. Instead, we derive a *conditional structural requirement*:

> **If** black hole formation and evaporation are unitary and information is recoverable from the full radiation record, **then** the compression–recovery theorem implies the existence of a compressed sector label K whose range controls the required side-information via $k \sim \log_2 |\mathrm{range}(K)|$.

This reframes the question from "is information preserved?" to "what minimal surviving sector structure must exist if information is preserved?"

4.2 Mapping to the compression–recovery framework

We map the black hole setting onto Theorem **??** as follows:

- **Microstate:** an initial pure state $|\Psi_0\rangle$ (equivalently, an interior microstate at formation).
- **Scrambling:** fast scrambling dynamics within the horizon and subsequent evaporation producing radiation with locally thermal statistics.
- **Local thermal appearance:** reduced states on limited subsets of Hawking quanta appear thermal, analogous to $\rho_B = I/2$ in teleportation or high-q scrambling in Hopfield.
- **Side information:** a compressed label K (sector/branch/code-subspace label) accessible from global constraints, boundary data, or asymptotic charges.
- **Decoder:** a reconstruction map that uses the full radiation record plus K to recover the correct sector and hence enable recovery within that sector.

The theorem does *not* assert which decoder quantum gravity provides; it asserts that *if* recovery is possible, some such $(K, \mathcal{D})$ pair must exist.

4.3 Conditional prediction: subexponential sector scaling

Let $\mathcal{H}$ denote the Hilbert space relevant to the evaporating black hole process, and suppose it admits a decomposition into distinguishable sectors indexed by K:

$$\mathcal{H} \approx \bigoplus_{K \in \mathrm{range}(K)} \mathcal{H}_K. \tag{1}$$

If recovery is possible after extreme scrambling, the compression–recovery theorem predicts that the minimal side-information required to select the correct sector scales as

$$k \sim \log_2 |\mathrm{range}(K)|. \tag{2}$$

The key scaling question is how $|\mathrm{range}(K)|$ grows relative to the black hole entropy S_{BH}:

- If $|\text{range}(K)| \sim e^{cS_{\text{BH}}}$, then $k = \Theta(S_{\text{BH}})$ and there is no compression (full microstate information is required).
- If $|\text{range}(K)| \sim \text{poly}(S_{\text{BH}})$, then $k = O(\log S_{\text{BH}})$ and compressed recovery is possible.
- If $|\text{range}(K)| = O(1)$, then $k = O(1)$ and maximal compression occurs.

Thus, a *falsifiable* conditional prediction is:

> **If unitarity holds and recovery is possible,** then the effective number of distinguishable recovery sectors must be subexponential in S_{BH}.

4.4 Candidate physical realizations of sector labels K

We now list mainstream structures in gravity/QFT that naturally provide sector labels compatible with the framework:

(i) Conserved asymptotic charges. In asymptotically flat spacetimes, ADM mass M, angular momentum J, and gauge charges Q label distinct superselection sectors. These quantities are, in principle, readable at infinity and constrain radiation kinematics.

(ii) Asymptotic symmetries and "soft hair." Large gauge transformations and diffeomorphisms at null infinity generate conserved soft charges, providing additional sector data beyond (M, J, Q).

(iii) Holographic boundary data (AdS/CFT). In holographic settings, boundary observables provide a complete description of bulk states within a code subspace. In the present framework, K can be interpreted as a coarse boundary label sufficient to select the relevant subspace/branch for reconstruction.

(iv) Page-branch / island-configuration labels. If the semiclassical entropy computation admits multiple saddle branches (e.g. "no-island" vs. "island"), the corresponding branch choice functions as a discrete sector label. More refined K can index families of island configurations.

4.5 Consistency with recent developments (evidence, not proof)

Recent progress in black hole information theory is *consistent* with the compression–recovery architecture, in the precise sense that it relies on global correlations and branch/sector structure to restore purification of radiation: (i) the Page curve behavior for unitary evaporation, (ii) quantum extremal surfaces and island formulas selecting entropy saddles, and (iii) holographic reconstruction via

boundary duals. These results do not prove the present framework, but they indicate that quantum gravity already employs the kind of global correlation structure the theorem requires.

4.6 Falsifiable predictions and failure modes

The framework yields sharp falsifiability criteria:

- **Exactly thermal late radiation:** If the full radiation record is exactly thermal with no recoverable long-range correlations, then either unitarity fails or the recovery channel is not in the radiation (e.g. remnants/baby universes), contradicting the simplest reconstruction picture.
- **Exponential sector proliferation:** If the number of required sectors scales as $|\mathrm{range}(K)| \sim e^{S_{\mathrm{BH}}}$, compression fails and side information must be extensive.
- **No decoder:** If no physically realizable decoder exists even given global constraints, then recoverability after scrambling is impossible in practice (and possibly in principle).

In summary: the theorem contributes a clean *conditional prediction*—if recovery is possible, sector labels must exist and must be low-dimensional relative to the microstate entropy.

4.7 Recoverability Implies a Low-Dimensional Sector Statistic

We formalize the minimal information-theoretic structure shared by the Hopfield and teleportation instantiations: recovery after scrambling requires a compressed label that selects the correct macro-sector.

Setup. Let X denote a microstate drawn from a finite ensemble $\mathcal{X}$ with prior $p(x)$. Let $Y = S(X)$ be the scrambled observation produced by a channel S (classical or quantum). Let $\alpha \in \mathcal{A}$ be the *sector index* (basin label, Bell outcome class, superselection label), with $\alpha = K(X)$ for some map $K : \mathcal{X} \to \mathcal{A}$.

A decoder is any map $\mathcal{D}$ producing an estimate $\widehat{\alpha}$ from accessible data:

$$\widehat{\alpha} = \mathcal{D}(Y, Z),$$

where Z is *side information* available to the decoder (e.g. clamped coordinates, a measurement transcript, boundary conditions).

Recoverability criterion. We say sector recovery is possible if the decoder achieves

$$\Pr[\widehat{\alpha} = \alpha] \geq 1 - \varepsilon.$$

Necessity of a compressed sector statistic. If recovery is possible, then (Y, Z) must contain nonzero mutual information about α:

$$I(\alpha; Y, Z) > 0.$$

By the data processing inequality, any recovery pipeline $\alpha \to X \to (Y, Z) \to \widehat{\alpha}$ satisfies

$$I(\alpha; \widehat{\alpha}) \leq I(\alpha; Y, Z).$$

If $\Pr[\widehat{\alpha} = \alpha] \geq 1 - \varepsilon$, then (standard bounds via Fano's inequality) imply that $I(\alpha; Y, Z)$ must be large enough to discriminate $|\mathcal{A}|$ sectors. In particular, the number of bits of side information required to specify the sector is lower-bounded by the log-cardinality:

$$k \quad \log_2 |\mathcal{A}| \quad (uptoadditivetermsdependingon\varepsilon).$$

Thus *recoverability implies existence of a low-dimensional statistic* K whose range size controls the side-information scaling.

Compression regime. Let S denote an effective entropy scale of the scrambled microstate family. If $|\mathcal{A}|$ grows subexponentially in S (e.g. polynomial), then

$$k = O(\log S)$$

and sector selection can be achieved with compressed side information. If $|\mathcal{A}| \sim e^{cS}$, then $k = O(S)$ and no compression occurs.

4.8 Scaling Constraint from Unitarity

Let a black hole of entropy S_{BH} have Hilbert space dimension

$$\dim(\mathcal{H}_{BH}) \sim e^{S_{BH}}.$$

Assume evaporation is unitary, so that the total state evolves as

$$U : \mathcal{H}_{BH} \to \mathcal{H}_{\text{rad}},$$

with U unitary.

Suppose information is scrambled during evaporation, so that local observables on early radiation appear thermal,

$$I(O_{\text{local}}; \alpha) \approx 0,$$

for sector index α.

For recovery to occur with compressed side information, there must exist a sector functional

$$K : \mathcal{H}_{BH} \to \mathcal{A},$$

such that:

1. K is invariant under scrambling dynamics.
2. Recovery fidelity depends only on K.

Let

$$M_{\text{sectors}} := |\mathcal{A}|.$$

By the sufficient-statistic lemma (Section 2), side information satisfies

$$k \sim \log_2 M_{\text{sectors}}.$$

If

$$M_{\text{sectors}} \sim e^{cS_{BH}},$$

then

$$k = O(S_{BH}),$$

and no compression occurs.

If instead

$$M_{\text{sectors}} = \text{poly}(S_{BH}),$$

then

$$k = O(\log S_{BH}),$$

and compression–recovery is feasible.

Scaling Requirement. If black hole evaporation is unitary and admits recovery from scrambled radiation, then the number of dynamically relevant sector labels must be subexponential in S_{BH}.

Interpretation. This subsection does not assume a specific physical realization of K. It states a *structural requirement*: whenever a system appears locally thermal after scrambling yet remains globally recoverable, there must exist a compressed sector statistic that survives in accessible correlations.

Relation to holography (preview). In AdS/CFT, bulk states are encoded in boundary degrees of freedom. If black-hole evaporation is unitary in such a holographic setting, then the accessible boundary data (or entanglement wedge observables) must contain sufficient information to select the correct bulk sector. In the language above, holography provides candidate realizations of (Y, Z) and, implicitly, of the sector statistic K. We emphasize that this does not derive AdS/CFT; it states what holographic recoverability would require in terms of sector scaling.

5 Field-Theoretic Instantiation: Compression–Recovery in a Coherence Field

5.1 State Space and Microstate Structure

Let $\mathcal{F}$ denote a coherence-bearing field with configuration space $\mathcal{X}$. A microstate is represented by a configuration

$$x \in \mathcal{X},$$

which may encode spatial, relational, or informational degrees of freedom.

We assume that coherent structures ("identities," attractors, or phase-locked configurations) correspond to dynamically stable subsets

$$\mathcal{B}_\alpha \subset \mathcal{X},$$

where α indexes a distinguishable sector.

The total state space decomposes as

$$\mathcal{X} = \bigcup_{\alpha \in \mathcal{A}} \mathcal{B}_\alpha,$$

with $\mathcal{A}$ the index set of coherent sectors.

5.2 Scrambling Dynamics

Let $\Phi_t : \mathcal{X} \to \mathcal{X}$ denote field evolution.

We define a scrambling regime as one in which:

- Local observables lose predictive power about α.
- The microstate is redistributed within a high-dimensional region of $\mathcal{X}$.

Formally, for a local observable O,

$$I(O(x_t); \alpha) \to 0$$

under scrambling evolution.

This parallels:

- Random spin flips in Hopfield systems,
- Local thermalization in black hole radiation.

5.3 Sector Label and Coherence Signature

Recovery requires existence of a sector label:

$$K : \mathcal{X} \to \mathcal{A}.$$

We define K as a *coherence signature functional* such that:

1. $K(x)$ is invariant under scrambling within sector α.
2. $K(x)$ distinguishes $\mathcal{B}_\alpha$ from $\mathcal{B}_\beta$.

Examples (abstractly defined):

- Spectral phase-lock signature
- Topological winding number
- Conserved global charge
- Relational boundary condition class

Let

$$M_{\text{sectors}} := |\mathcal{A}|.$$

Side-information scaling then satisfies

$$k \sim \log_2 M_{\text{sectors}}.$$

5.4 Consistency with Page Curve Behavior

The Page curve describes the entanglement entropy of Hawking radiation as a function of evaporation time.

For a black hole formed in a pure state, the radiation entropy $S_{\text{rad}}(t)$ behaves schematically as:

$$S_{\text{rad}}(t) \sim \{\ increasing, t < t_{\text{Page}}, decreasing, t > t_{\text{Page}}.$$

The early-time growth reflects effective thermal behavior, where local observables reveal no information about the initial state.

The late-time decrease implies that correlations between early and late radiation restore purity.

Interpretation within Compression–Recovery Framework. Early-time regime:

- Scrambling dominates.
- Local measurements carry negligible sector information.
- Radiation appears thermal.

Late-time regime:

- Global correlations become accessible.
- Sector information becomes reconstructible.
- Effective decoder behavior emerges.

Thus the Page transition corresponds structurally to:

$$Scramblingphase \longrightarrow Sector - resolutionphase.$$

This behavior is consistent with the existence of a sector functional K whose information is nonlocally encoded in radiation degrees of freedom.

Important Caveat. The Page curve alone does not prove that sector labels exist, nor does it establish their scaling behavior. However, the existence of a Page transition is compatible with a compression–recovery interpretation of black hole evaporation.

5.5 Decoder as Resonant Convergence Operator

Define a decoder map

$$\mathcal{D}_K : \mathcal{X} \to \mathcal{B}_{K(x)}.$$

The decoder acts via gradient flow on a coherence functional

$$\mathcal{C}(x),$$

such that

$$\frac{dx}{d\tau} = -\nabla \mathcal{C}(x),$$

subject to boundary condition determined by K.

This parallels:

- Energy descent in Hopfield networks,
- Conditional unitary in teleportation,
- Entanglement wedge reconstruction in holography.

5.6 Scaling Requirement

For recovery to occur with compressed side information,

$$M_{\text{sectors}} must be subexponential in effective entropy scale S.$$

If

$$M_{\text{sectors}} \sim e^{cS},$$

then

$$k = O(S),$$

and no compression occurs.

If

$$M_{\text{sectors}} \sim \text{poly}(S),$$

then

$$k = O(\log S),$$

and compression–recovery is feasible.

5.7 Falsifiability

The framework fails if:

1. No invariant sector functional K exists.
2. M_{sectors} scales exponentially in entropy.
3. No convergent decoder flow exists.

The claim is not that all coherence fields obey this structure, but that any recoverable coherence field must.

6 Field-Theoretic Instantiation: Compression–Recovery in a Coherence Field

6.1 State Space and Microstate Structure

Let $\mathcal{F}$ denote a coherence-bearing field with configuration space $\mathcal{X}$. A microstate is represented by a configuration

$$x \in \mathcal{X},$$

which may encode spatial, relational, or informational degrees of freedom.

We assume that coherent structures ("identities," attractors, or phase-locked configurations) correspond to dynamically stable subsets

$$\mathcal{B}_\alpha \subset \mathcal{X},$$

where α indexes a distinguishable sector.

The total state space decomposes as

$$\mathcal{X} = \bigcup_{\alpha \in \mathcal{A}} \mathcal{B}_\alpha,$$

with $\mathcal{A}$ the index set of coherent sectors.

6.2 Scrambling Dynamics

Let $\Phi_t : \mathcal{X} \to \mathcal{X}$ denote field evolution.

We define a scrambling regime as one in which:

- Local observables lose predictive power about α.
- The microstate is redistributed within a high-dimensional region of $\mathcal{X}$.

Formally, for a local observable O,

$$I(O(x_t); \alpha) \to 0$$

under scrambling evolution.

This parallels:

- Random spin flips in Hopfield systems,
- Local thermalization in black hole radiation,
- Projection-induced mixing in quantum protocols.

6.3 Sector Label and Coherence Signature

Recovery requires existence of a sector label:

$$K : \mathcal{X} \to \mathcal{A}.$$

We define K as a *coherence signature functional* such that:

1. $K(x)$ is invariant under scrambling within sector α.
2. $K(x)$ distinguishes $\mathcal{B}_\alpha$ from $\mathcal{B}_\beta$.

In URF terminology, K may be instantiated by a *braid key*: a minimal continuity token that does not encode the full microstate x, but provides sufficient side-information to constrain decoding flow into the correct basin $\mathcal{B}_\alpha$ after scrambling.

Let

$$M_{\text{sectors}} := |\mathcal{A}|.$$

Side-information scaling then satisfies

$$k \sim \log_2 M_{\text{sectors}}.$$

6.4 Decoder as Resonant Convergence Operator

Define a decoder map

$$\mathcal{D}_K : \mathcal{X} \to \mathcal{B}_{K(x)}.$$

The decoder acts via gradient flow on a coherence functional

$$\mathcal{C}(x),$$

such that

$$\frac{dx}{d\tau} = -\nabla \mathcal{C}(x),$$

subject to boundary conditions determined by K.

This parallels:

- Energy descent in Hopfield networks,
- Conditional unitary in teleportation,
- Entanglement wedge reconstruction in holography.

6.4.1 Example: Gravity as a Coherence-Gradient Functional

As one concrete instantiation of the abstract coherence functional $\mathcal{C}(x)$, prior URF work models gravitational potential as proportional to the gradient of an integrated coherence content C:

$$\nabla\Phi(x,t) = -\frac{G}{c^2}\nabla C(x,t), \qquad C(x,t) = \int |\Psi_{\text{res}}(x,t)|^2\, dV.$$

In this formulation, effective mass corresponds to condensed coherence density,

$$M = \frac{1}{c^2}\int \rho_{\text{coh}}(x,t)\, dV, \qquad \rho_{\text{coh}} = \Lambda_{\text{URF}}|\Psi_{\text{res}}|^2.$$

This provides an explicit candidate for $\mathcal{C}(x)$ and its induced decoder flow. We emphasize that Section 6 requires only the *existence* of such a functional for recovery; this example is provided as a worked model, not as an established claim about gravity.

6.5 Scaling Requirement

For recovery to occur with compressed side information,

$$M_{\text{sectors}} mustbesubexponentialineffectiveentropyscaleS.$$

If

$$M_{\text{sectors}} \sim e^{cS},$$

then

$$k = O(S),$$

and no compression occurs.

If

$$M_{\text{sectors}} \sim \text{poly}(S),$$

then

$$k = O(\log S),$$

and compression–recovery is feasible.

6.6 Falsifiability

The framework fails if:

1. No invariant sector functional K exists.
2. M_{sectors} scales exponentially in entropy.
3. No convergent decoder flow exists.

The claim is not that all coherence fields obey this structure, but that any recoverable coherence field must.

A Hopfield Network Simulations (Numerical Validation)

A.1 Model and Stored Patterns

We validate the compression–recovery framework using a classical Hopfield network with N binary spins $s_i \in \{-1, +1\}$ and M stored patterns $\{\xi^\mu\}_{\mu=1}^M$, where $\xi^\mu \in \{-1, +1\}^N$ are drawn i.i.d. uniformly at random.

The (Hebbian) coupling matrix is

$$J_{ij} = \frac{1}{N} \sum_{\mu=1}^{M} \xi_i^\mu \xi_j^\mu, \qquad J_{ii} = 0. \tag{3}$$

The Hopfield energy is

$$E(\mathbf{s}) = -\frac{1}{2} \sum_{i \neq j} J_{ij}\, s_i s_j. \tag{4}$$

A.2 Order Parameters and Success Metric

For any state $\mathbf{s}$ and pattern ξ^μ, define the overlap

$$m^\mu(\mathbf{s}) = \frac{1}{N} \sum_{i=1}^{N} \xi_i^\mu s_i \in [-1, 1]. \tag{5}$$

Given a designated target pattern index $\mu^\star$, we say reconstruction is *successful* if

$$\arg \max_{\mu \in \{1,\ldots,M\}} m^\mu(\mathbf{s}_{\text{final}}) = \mu^\star, \tag{6}$$

and report the *target fidelity* as $m^{\mu^\star}(\mathbf{s}_{\text{final}})$.

A.3 Scrambling and Side Information

Each trial proceeds as follows:

(1) Storage and target selection. Generate M random patterns, construct J, and select a target index $\mu^\star$ uniformly.

(2) Scrambling. Initialize a fully scrambled state $\mathbf{s}_0$ by drawing each $s_{0,i}$ uniformly from $\{-1, +1\}$ independently. This produces a high-entropy microstate with overlaps concentrated near zero.

(3) Side information via clamping. To model a compressed sector label K in the discrete Hopfield setting, we clamp a subset $I \subset \{1, \ldots, N\}$ of size b to match the target pattern:

$$s_i \leftarrow \xi_i^{\mu^\star} \quad forall i \in I. \tag{7}$$

We parameterize the clamp size by the fraction $q = b/N \in [0, 1]$. This is the Hopfield analogue of providing k bits of side information to restrict the state to a constraint manifold that contains the target basin but excludes most competing basins.

A.4 Dynamics (Asynchronous Updates)

We evolve the state with asynchronous Glauber-style updates for T sweeps (one sweep $\approx N$ attempted updates). For each update, choose $i \sim \mathrm{Unif}\{1, \ldots, N\}$ and compute the local field

$$h_i(\mathbf{s}) = \sum_{j=1}^{N} J_{ij}\, s_j. \tag{8}$$

At inverse temperature $\beta > 0$, the stochastic update is

$$P(s_i \leftarrow +1) = \frac{1}{1 + \exp(-2\beta h_i)}, \qquad P(s_i \leftarrow -1) = 1 - P(s_i \leftarrow +1). \tag{9}$$

Clamped coordinates are held fixed (updates at $i \in I$ are skipped).

A.5 Experimental Grid and Reported Quantities

We sweep the load parameter

$$\alpha = \frac{M}{N} \tag{10}$$

and clamping fraction $q = b/N$ over a grid. For each (α, q) pair we run R independent trials, and report:

- **Success probability:** $\hat{p}_{\mathrm{succ}}(\alpha, q)$ = fraction of trials with correct basin recovery.
- **Mean target overlap:** $\widehat{E}[m^{\mu^\star}(\mathbf{s}_{\mathrm{final}})]$ as a fidelity proxy.

A.6 Results: Phase Diagram and Fidelity Landscape

Figure 1 shows the phase diagram of reconstruction success probability as a function of (α, q). As expected, recovery degrades as α approaches the classical capacity regime, and improves sharply with increased q (more side information). Figure 2 shows the corresponding mean target overlap.

fig_A1_phase_diagram.png

Figure 1: Hopfield reconstruction success probability $\hat{p}_{\mathrm{succ}}(\alpha, q)$ across a grid of load $\alpha = M/N$ and clamping fraction $q = b/N$. Each cell is estimated from R independent trials. A sharp transition emerges: modest side information q yields near-certain basin recovery in the sub-capacity regime, while failure occurs when the attractor landscape is overloaded (large α) or side information is insufficient (small q).

A.7 Interpretation in the Compression–Recovery Framework

These simulations operationalize the theorem's structure in a discrete, fully replicable setting:

- **Scrambling:** $\mathbf{s}_0$ is maximally disordered; local observables appear thermal.
- **Side information:** the clamping set I restricts the state to a constraint manifold $\mathcal{M}_I$ that contains the target basin by construction and probabilistically excludes most competing basins.
- **Decoder:** Hopfield dynamics performs gradient-like descent in the energy

fig_A2_fidelity.png

Figure 2: Mean target overlap (fidelity proxy) $\widehat{E}[m^{\mu^\star}(\mathbf{s}_{\text{final}})]$ across the same (α, q) grid. High overlap correlates with correct basin identification; the overlap collapses near the failure boundary visible in Fig. 1.

landscape, converging to the compatible attractor.

Empirically, the success boundary is consistent with the claim that recovery is controlled by a sublinear side-information knob: increasing the constraint strength (here, q) produces a rapid sigmoidal transition from failure to reliable basin selection.

A Hopfield Network Simulations (Replication Appendix)

This appendix provides a replication-ready numerical validation of the Hopfield instantiation used in the main text. The goal is not to discover new Hopfield phenomena, but to provide concrete evidence for the compression–recovery mechanism and its phase boundary as a function of (i) memory load and (ii) scrambling.

A.1 Model and Storage Rule

We consider a standard Hopfield network with N binary spins $s_i \in \{-1, +1\}$ and P stored patterns $\{\xi^\mu\}_{\mu=1}^P$, where each pattern $\xi^\mu \in \{-1, +1\}^N$ is drawn i.i.d. uniformly at random. The Hebbian weight matrix is

$$J_{ij} = \frac{1}{N} \sum_{\mu=1}^{P} \xi_i^\mu \xi_j^\mu, \qquad J_{ii} = 0. \tag{11}$$

The (zero-temperature) Hopfield energy is

$$E(\mathbf{s}) = -\frac{1}{2} \sum_{i \neq j} J_{ij} s_i s_j. \tag{12}$$

Define the load parameter $\alpha := P/N$.

A.2 Scrambling and Retrieval Protocol

For each trial:

1. Generate $P = \lfloor \alpha N \rfloor$ random patterns and construct J.
2. Choose a target pattern index $\mu^\star \in \{1, \ldots, P\}$ uniformly.
3. Initialize state $\mathbf{s}$ at the target: $\mathbf{s} \leftarrow \xi^{\mu^\star}$.
4. Scramble by flipping each spin independently with probability q:
$$s_i \leftarrow \{ - s_i withprob.q, \quad s_i withprob.1 - q.$$
5. Run asynchronous zero-temperature updates (Glauber dynamics): for a random index i,
$$h_i = \sum_j J_{ij} s_j, \qquad s_i \leftarrow \mathrm{sign}(h_i),$$
breaking ties randomly if $h_i = 0$.
6. After convergence or a maximum of $T_{\max}$ updates, compute overlaps
$$m^\mu(\mathbf{s}) = \frac{1}{N} \sum_i \xi_i^\mu s_i,$$
and declare success if $\arg\max_\mu m^\mu = \mu^\star$ and $m^{\mu^\star}$ exceeds a fixed threshold.

A.3 Outputs and Figures

The replication script generates:

- A phase diagram of success probability over a grid of (α, q) values.
- A fidelity plot showing mean recovered overlap $m^{\mu^\star}$ across the same grid.

fig_A1_demo_phase.png

Figure 3: Demo phase diagram (small N). Color indicates empirical success probability over (α, q).

A.4 Replication Instructions

Run the script (Python 3.9+ recommended):

```
python hopfield_sim.py --N 400 --trials 200 --out hopfield_publication
```

This produces:

```
hopfield_publication_results.csv
fig_A1_phase_diagram.png
fig_A2_fidelity.png
```

Increase N and trials for smoother boundaries. !/usr/bin/env python3 """ Hopfield compression–recovery simulations (Appendix A).

This script generates: - Phase diagram: success probability vs (alpha=P/N, q scrambling rate) - Fidelity diagram: mean recovered overlap m$_{mu*}vs(alpha, q)$

Outputs: - CSV with grid results - PNG figures suitable for inclusion in a paper

Usage example: python hopfield$_{s}im.py--N400--trials200--outhopfield_{p}ublication$

fig_A2_demo_fidelity.png

Figure 4: Demo fidelity plot (small N). Color indicates mean recovered overlap $m^{\mu^\star}$.

Notes: - This is a numerical validation appendix, not an optimized solver. - For publication figures, increase N and trials. """

import argparse import csv import os import numpy as np import matplotlib.pyplot as plt

def generate$_p atterns(P : int, N : int, rng : np.random.Generator)- > np.ndarray : """Randomi.i.d.patternsin-1, +1^{PxN}."""returnrng.choice([-1, 1], size = (P, N))$

def hopfield$_w eights(patterns : np.ndarray)- > np.ndarray : """HebbianweightsJ_ij = (1/N)sum_m uxi_i^m uxi_j^m uwithzerodiagonal."""P, N = patterns.shapeJ = (patterns.T@patterns)/Nnp.fill_d iag$

def scramble$_s tate(s : np.ndarray, q : float, rng : np.random.Generator)- > np.ndarray : """Flipeachspinindependentlywithprobabilityq."""flips = rng.random(size = s.shape[0]) < qs_n ew = s.copy()s_n ew[flips]* = -1returns_n ew$

def async$_z ero_t emp_d ynamics(J : np.ndarray, s : np.ndarray, steps : int, rng : np.random.Generator)- > np.ndarray : """Asynchronouszero-temperatureGlauberdynamics."""N = s.shape[0]for_i nrange(steps) : i = rng.integers(0, N)h = J[i]@sifh > 0 : s[i] = 1elifh < 0 : s[i] = -1else : s[i] = rng.choice([-1, 1])returns$

def overlaps(patterns: np.ndarray, s: np.ndarray) -¿ np.ndarray: """m$^m u =$

$(1/N)sum_i xi_i^m us_i."""N = s.shape[0]return(patterns@s)/N$

def run$_t rial(N : int, alpha : float, q : float, rng : np.random.Generator, max_s weeps :$
$int = 5, success_t hresh : float = 0.8) - > tuple[bool, float] : """Runonescrambleßrecovertrialandreturn(succe$
$max(1, int(round(alpha*N)))pats = generate_p atterns(P, N, rng)J = hopfield_w eights(pats)$

mu$_s tar = rng.integers(0, P)s = pats[mu_s tar].copy()s = scramble_s tate(s, q, rng)$

s = async$_z ero_t emp_d ynamics(J, s, steps = max_s weeps * N, rng = rng)$

m = overlaps(pats, s) mu$_h at = int(np.argmax(m))fidelity = float(m[mu_s tar])success =$
$(mu_h at == mu_s tar)and(fidelity >= success_t hresh)returnsuccess, fidelity$

def grid$_s weep(N : int, alphas : np.ndarray, qs : np.ndarray, trials : int, seed :$
$int, max_s weeps : int = 5, success_t hresh : float = 0.8) : rng = np.random.default_r ng(seed)success_g rid =$
$np.zeros((len(alphas), len(qs)))fidelity_g rid = np.zeros((len(alphas), len(qs)))fidelity_s td =$
$np.zeros((len(alphas), len(qs)))$

for i, a in enumerate(alphas): for j, q in enumerate(qs): successes = [] fidelities = [] for $_i nrange(trials) : ok, fid = run_t rial(N, a, q, rng, max_s weeps =$
$max_s weeps, success_t hresh = success_t hresh)successes.append(1.0ifokelse0.0)fidelities.append(fid)success_g$
$float(np.mean(successes))fidelity_g rid[i, j] = float(np.mean(fidelities))fidelity_s td[i, j] =$
$float(np.std(fidelities))returnsuccess_g rid, fidelity_g rid, fidelity_s td$

def save$_c sv(outpath : str, alphas : np.ndarray, qs : np.ndarray, success_g rid :$
$np.ndarray, fidelity_g rid : np.ndarray, fidelity_s td : np.ndarray) : withopen(outpath, "w", newline =$
$"")asf : w = csv.writer(f)w.writerow(["alpha", "q", "success_p rob", "mean_f idelity", "std_f idelity"])fori, aine$
$forj, qinenumerate(qs) : w.writerow([a, q, success_g rid[i, j], fidelity_g rid[i, j], fidelity_s td[i, j]])$

def plot$_h eatmap(data : np.ndarray, alphas : np.ndarray, qs : np.ndarray, title :$
$str, outpng : str, xlabel : str = r"$q(scrambling)", ylabel : str = r"\alpha = P/N"):$
plt.figure() plt.imshow(data, origin="lower", aspect="auto", extent=[qs.min(), qs.max(), alphas.min(), alphas.max()], interpolation="nearest",) plt.colorbar() plt.xlabel(xlabel) plt.ylabel(ylabel) plt.title(title) plt.tight$_l ayout()plt.savefig(outpng, dpi =$
$250)plt.close()$

def main(): ap = argparse.ArgumentParser() ap.add$_a rgument("--N", type =$
$int, default = 50)ap.add_a rgument("--trials", type = int, default = 50)ap.add_a rgument("-$
$-alpha_m in", type = float, default = 0.02)ap.add_a rgument("--alpha_m ax", type =$
$float, default = 0.18)ap.add_a rgument("--alpha_s teps", type = int, default =$
$10)ap.add_a rgument("--q_m in", type = float, default = 0.0)ap.add_a rgument("-$
$-q_m ax", type = float, default = 0.5)ap.add_a rgument("--q_s teps", type =$
$int, default = 11)ap.add_a rgument("--max_s weeps", type = int, default =$
$5)ap.add_a rgument("--success_t hresh", type = float, default = 0.8)ap.add_a rgument("-$
$-seed", type = int, default = 0)ap.add_a rgument("--out", type = str, default =$
$"demo")args = ap.parse_a rgs()$

alphas = np.linspace(args.alpha$_m in, args.alpha_m ax, args.alpha_s teps)qs =$
$np.linspace(args.q_m in, args.q_m ax, args.q_s teps)$

success$_g rid, fidelity_g rid, fidelity_s td = grid_s weep(N = args.N, alphas =$
$alphas, qs = qs, trials = args.trials, seed = args.seed, max_s weeps = args.max_s weeps, success_t hresh =$
$args.success_t hresh,)$

os.makedirs(os.path.dirname(args.out), exist$_o k = True)ifos.path.dirname(args.out)elseNonecsv_p ath =$
$f"args.out_r esults.csv"save_c sv(csv_p ath, alphas, qs, success_g rid, fidelity_g rid, fidelity_s td)$

plot$_h eatmap(success_g rid, alphas, qs, title = f"HopfieldRecoverySuccess(N =$
$args.N, trials = args.trials)", outpng = f"fig_A 1_{os.path.basename(args.out)p} hase.png",)plot_h eatmap(fidelity_g ri$

$f"HopfieldRecoveryFidelity(N = args.N, trials = args.trials)", outpng =$
$f"fig_A2_{os.path.basename(args.out)}fidelity.png",)$

print("Wrote:", csv$_{p}ath)print("Wrotefigures : fig_A1_{..}_{p}hase.png, fig_A2_{..}_{f}idelity.png")$

if $_{n}ame_{=="}{}_{m}ain_{":main()to}$